Nikhil Bindal

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Professional Summary

Full-Stack & AI Infrastructure Engineer with 6+ years of experience building distributed systems, real-time streaming platforms, and agentic AI applications across fintech, media, research, and AI startups. Experienced in designing low-latency AI workflows, multimodal RAG systems, orchestration pipelines, knowledge graphs, and production-grade backend architectures using Python, FastAPI, Node.js, PostgreSQL, Redis, and cloud infrastructure.

Skills

Backend & APIs	Python (FastAPI, Django, AsyncIO), Node.js (Express), REST APIs, WebSockets, gRPC
AI / LLM	Agentic Workflows, RAG Systems, LangGraph, CrewAI, LlamaIndex, GPT-4o, Gemini, Vertex AI
Databases	PostgreSQL, MongoDB, Redis, Neo4j, Pinecone, MySQL
Infrastructure	AWS, GCP, Docker, Kubernetes, Prometheus, Grafana, SQS
Architecture	Distributed Systems, Event-Driven Systems, Real-Time Streaming, Caching, Async Processing
Frontend	React.js, React Native, TypeScript

Experience

AI Consultant / AI Infrastructure Engineer

San Francisco, CA | Sep 2024 – Present

Neurologica (Presently) — Engineering Contractor

- Architected and built a production-grade AI coaching platform combining biometric signal analysis with a multi-agent orchestration pipeline and persistent personal memory systems.
- Designed and implemented a 6-agent orchestration pipeline reducing end-to-end AI latency from 36s to 9.5s (73% reduction) through prompt consolidation, model tiering, and async execution optimization.
- Built Mnemosyne memory architecture with semantic retrieval, temporal decay lifecycle, and context-scoped memory bucketing across multiple user domains.
- Engineered Redis-backed concurrency controls and session orchestration preventing state corruption under concurrent real-time voice sessions.
- Built parallel voice sidecar architecture using Gemini Live enabling low-latency audio conversations while asynchronously running sentiment, query, and analytics pipelines.
- Integrated real-time HCI signal processing workflows using multimodal emotion/context signals to adapt conversational reasoning dynamically.
- Developed Python and JavaScript SDKs supporting 32 operations across orchestration, memory, analytics, sessions, and BYOD infrastructure.
- Established CI/CD pipelines, audit logging infrastructure, webhook delivery systems, and production deployment workflows on GCP Cloud Run.

Stealth AI Startup — AI Solutions Consultant

- Architected an AI-powered meeting intelligence platform automating pre-meeting research, live meeting assistance, and post-meeting synthesis workflows.
- Designed a 3-phase agentic architecture orchestrating 15 specialized AI agents across research, meeting intelligence, and decision-tracking pipelines.
- Built contextual retrieval workflows using multi-level semantic search across Qdrant vector collections improving retrieval relevance by ~40% over traditional RAG pipelines.
- Implemented a real-time voice pipeline using LiveKit WebRTC, Deepgram STT, and Cartesia TTS targeting sub-200ms perceived latency.
- Designed PostgreSQL schemas and knowledge graph workflows modeling meeting entities, action items, decisions, transcripts, and research relationships.

- Implemented Redis-backed rate limiting, Prometheus metrics, structured logging, and async background workers for scalable production monitoring.

Full-Stack Developer / Research Assistant — Northeastern University

Boston, MA | Feb 2023 – Dec 2023

- Developed a semantic search and data-visualization platform for biomedical research using FastAPI, PostgreSQL, React, and D3.js, enabling researchers to search and analyze 10K+ scientific papers and molecular datasets.
- Built AWS Batch-based distributed simulation pipelines orchestrating large-scale molecular computations and reducing compute costs by 40%.
- Led backend development for scientific data ingestion pipelines, metadata APIs, and ML inference workflows supporting computational biology research.
- Designed accessibility-focused tooling including tactile graphics generation and screen-reader integrations to improve STEM accessibility for visually impaired students and researchers.

Software Engineer — Times Internet

Noida, India | Apr 2021 – Jul 2022

- Built and scaled backend services for TOI+ subscription platform handling millions of daily requests.
- Migrated 70+ city portals from legacy systems to React-based micro-frontends improving performance and maintainability.
- Improved personalization and recommendation workflows supporting higher user engagement and retention.

Founding Software Engineer — Progcap

New Delhi, India | Jan 2019 – Mar 2021

- Built real-time lending and underwriting microservices using Node.js, Kafka, MongoDB, and event-driven architecture.
- Reduced transaction workflow latency from 8.7s to 890ms while supporting high-throughput lending operations.
- Integrated ML-powered credit scoring workflows into backend lending systems for large-scale financial operations.

Software Engineer — LiveMedia / LiveChek

New Delhi, India | Aug 2017 – May 2018

- Worked on insurance-focused telematics and behavioral analytics systems integrating real-time driving behavior and user interaction data.
- Built backend services and APIs for processing user activity streams and contextual insurance-related workflows.
- Contributed to early-stage product development and architecture decisions in a startup environment.

Selected Projects

- **Gaussian Splatting Knowledge Graph:** Designed a multi-agent AI pipeline that ingests academic PDFs, semantically chunks documents, extracts entities/relationships, resolves canonical duplicates, and constructs a queryable research knowledge graph.
- **RecoMe — Personalized Recommendation Engine:** Built a behavioral recommendation system that models user preferences and interaction patterns across platforms to personalize content discovery workflows.
- **Voice Document Intelligence System:** Built a real-time AI document assistant supporting voice-based contextual Q&A over uploaded documents using FastAPI, WebRTC, Redis, and multi-agent orchestration.

Education

- M.Sc. in Artificial Intelligence — University of the Cumberlands (2024 – 2025)
- M.Sc. in Information Systems — Northeastern University (2022 – 2024)
- B.Tech. in Computer Science & Engineering — Kurukshetra University (2012 – 2016)